References:

- Chapter 6 of Saville and Wood, *Statistical Methods: A Geometric Approach*.

We continue from last time.

How much does fertilizer affect yield? We record the two groups:

Treatment	y
1	29.80
1	28.57
1	29.97
2	30.33
2	31.27
2	30.37

One-hot / dummy variable encoding of the treatment levels:

Treatment	Value	x_1	x_2
1	$y_{11} = 29.80$	1	0
1	$y_{12} = 28.57$	1	0
1	$y_{13} = 29.97$	1	0
2	$y_{21} = 30.33$	0	1
2	$y_{22} = 31.27$	0	1
2	$y_{23} = 30.37$	0	1

Treatment	Value	x_1	x_2
1	$y_{11} = 29.80$	1	0
1	$y_{12} = 28.57$	1	0
1	$y_{13} = 29.97$	1	0
2	$y_{21} = 30.33$	0	1
2	$y_{22} = 31.27$	0	1
2	$y_{23} = 30.37$	0	1

Model: We can write the model in a few ways:

$$y = \begin{cases} \mu_1 & \text{if } y \text{ is from treatment } 1 \\ \mu_2 & \text{if } y \text{ is from treatment } 2 \end{cases}$$

As a homogeneous equation (one-hot encoding):

$$y = \mu_1 x_1 + \mu_2 x_2$$

As an inhomogeneous equation (dummy variable encoding with treatment 2 as reference level):

$$y = (\mu_1 - \mu_2) x_1 + \mu_2$$

As an inhomogeneous equation (dummy variable encoding with treatment 1 as reference level):

$$y = \mu_1 + (\mu_2 - \mu_1) x_2$$

Fitted model:

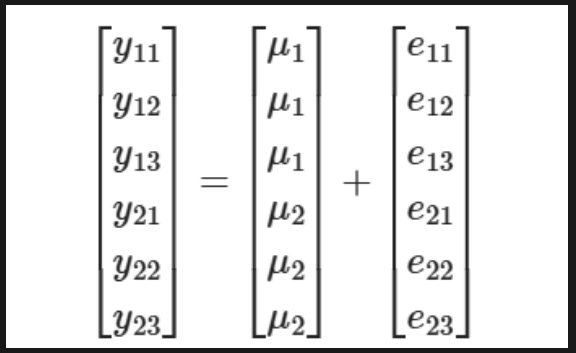
$$\begin{bmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{21} \\ y_{22} \\ y_{23} \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_1 \\ \mu_1 \\ \mu_2 \\ \mu_2 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} e_{11} \\ e_{12} \\ e_{13} \\ e_{21} \\ e_{22} \\ e_{23} \end{bmatrix}$$

$$\mathrm{df}_{\mathrm{Total}} = ?$$

$$\mathrm{df}_{\mathrm{Model}} = ?$$

$$\mathrm{df}_{\mathrm{Error}} = ?$$

Fitted model:



$$\begin{bmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{21} \\ y_{22} \\ y_{23} \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_1 \\ \mu_1 \\ \mu_2 \\ \mu_2 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} e_{11} \\ e_{12} \\ e_{13} \\ e_{21} \\ e_{22} \\ e_{23} \end{bmatrix}$$

$$\mathrm{df}_{\mathrm{Total}} = 6.$$

$$\mathrm{df}_{\mathrm{Model}} = 2.$$

$$\mathrm{df}_{\mathrm{Error}} = 6 - 2 = 4.$$

Fitted model (with actual values):

$$\underbrace{\begin{bmatrix} 29.80 \\ 28.57 \\ 29.97 \\ 30.33 \\ 31.27 \\ 30.37 \end{bmatrix}}_{\text{Observation vector}} = \underbrace{\begin{bmatrix} 29.447 \\ 29.447 \\ 29.447 \\ 30.657 \\ 30.657 \\ 30.657 \end{bmatrix}}_{\text{Fitted model vector}} + \underbrace{\begin{bmatrix} .353 \\ -.877 \\ .523 \\ -.327 \\ .613 \\ -.287 \end{bmatrix}}_{\text{Fitted error vector}}$$

Hypothesis testing (preliminary): Our model space $(\mu_1, \mu_1, \mu_1, \mu_2, \mu_2, \mu_2)^T$ is spanned by:

Hypothesis testing (preliminary): Our model space $(\mu_1, \mu_1, \mu_1, \mu_2, \mu_2, \mu_2)^T$ is spanned by:

- the vector $(1, 1, 1, 0, 0, 0)^T$ which tests the hypothesis $H_0 : \mu_1 = 0$, and
- the vector $(0, 0, 0, 1, 1, 1)^T$ which tests the hypothesis $H_0 : \mu_2 = 0$.

Hypothesis testing: Also by

- $(1, 1, 1, -1, -1, -1)^T$ which tests $H_0 : \mu_1 = \mu_2$ or $\mu_1 - \mu_2 = 0$.
- $(1, 1, 1, 1, 1, 1)^T$ which tests $H_0 : \mu = 0$ where μ is **grand mean** or **overall mean**, estimated by mean $\bar{y}_{\bullet,\bullet}$ of all observations.

Refined orthogonal decomposition:

$$y = \bar{y}_{..} + (\bar{y}_{i\cdot} - \bar{y}_{..}) + (y - \bar{y}_{i\cdot})$$

$$\underbrace{\begin{bmatrix} 29.80 \\ 28.57 \\ 29.97 \\ 30.33 \\ 31.27 \\ 30.37 \end{bmatrix}}_{\text{Observation vector}} = \underbrace{\underbrace{\begin{bmatrix} 30.052 \\ 30.052 \\ 30.052 \\ 30.052 \\ 30.052 \\ 30.052 \end{bmatrix}}_{\text{Overall mean vector}} + \underbrace{\begin{bmatrix} -.605 \\ -.605 \\ -.605 \\ .605 \\ .605 \\ .605 \end{bmatrix}}_{\text{Treatment vector}}}_{\text{Model}} + \underbrace{\begin{bmatrix} .353 \\ -.877 \\ .523 \\ -.327 \\ .613 \\ -.287 \end{bmatrix}}_{\text{Fitted error vector}}$$

ANOVA table:

Source of Variation	df	SS	MS	F
Overall mean ($H_0 : \mu = 0$)	1	5418.6160	5418.6160	
Treatments ($H_0 : \mu_1 = \mu_2$)	1	2.1962	2.1962	5.07 (ns)
Error	4	1.7324	0.4331	
Total	6	5422.5445		

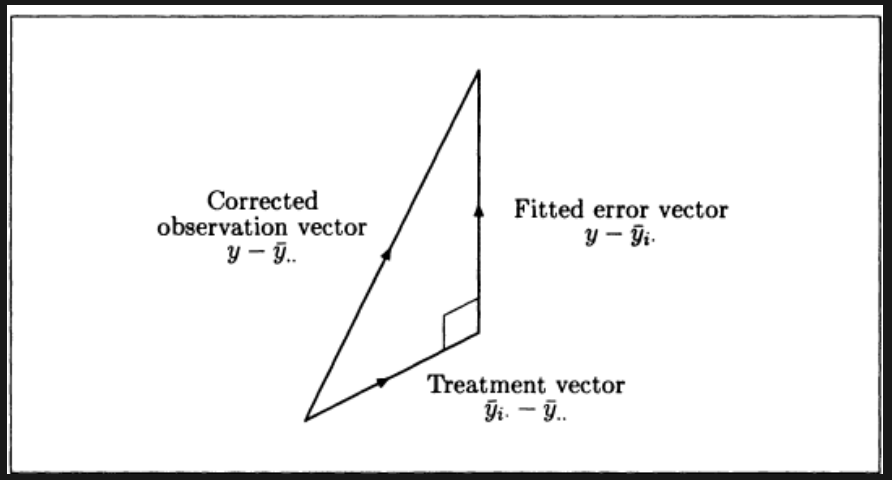
Table 6.2: ANOVA table for our two treatment bulkometer example.

Simplified decomposition:

$$y - \bar{y}_{..} = (\bar{y}_{i\cdot} - \bar{y}_{..}) + (y - \bar{y}_{i\cdot})$$

$$\begin{bmatrix} -.252 \\ -1.482 \\ -.082 \\ .278 \\ 1.218 \\ .318 \end{bmatrix} = \begin{bmatrix} -.605 \\ -.605 \\ -.605 \\ .605 \\ .605 \\ .605 \end{bmatrix} + \begin{bmatrix} .353 \\ -.877 \\ .523 \\ -.327 \\ .613 \\ -.287 \end{bmatrix}$$

Simplified decomposition:



Simplified ANOVA table:

Source of Variation	df	SS	MS	F
Treatments ($H_0 : \mu_1 = \mu_2$)	1	2.1962	2.1962	5.07 (ns)
Error	4	1.7324	.4331	
Total (corrected)	5	3.9286		

Table 6.3: The more usual form of the ANOVA table for the two treatment bulkometer example.

General case:

- Group 1 with n_1-many values $y_{11}, y_{12}, \ldots, y_{1n_1}$;
- Group 2 with n_2-many values $y_{21}, y_{22}, \ldots, y_{2n_2}$;

Then:

Source of Variation	df	SS	MS
Treatments	1	$\|\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}\|^2$	$\|\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}\|^2$
Error	$n_1 + n_2 - 2$	$\|y - \bar{y}_{i\cdot}\|^2$	$\frac{\|y - \bar{y}_{i\cdot}\|}{n_1 + n_2 -}$
Total (corrected)	$n_1 + n_2 - 1$	$\|y - \bar{y}_{\cdot\cdot}\|^2$	

Assumptions: This model assumes each group has equal variance. If this assumption fails, then use a more general form of ANOVA called Welch's ANOVA (like Welch's t-test).

SAS IMPLEMENTATION

Using PROC REG: Use dummy variables.

```
DATA study;
    INPUT Treatment y x1 x2;
    DATALINES;
1 29.80 1 0
1 28.57 1 0
1 29.97 1 0
2 30.33 0 1
2 31.27 0 1
2 30.37 0 1
;
RUN;
```

Using PROC REG: Use dummy variables.

```
/* Test H0: mu_1 = mu_2 using PROC REG */
PROC REG DATA=study;
    /* Model using x1 as dummy variable */
    MODEL y = x1;

    /* Test H0: mu_1 - mu_2 = 0 */
    Test_Mu1_vs_Mu2: TEST x1 = 0;
RUN;
QUIT;
```

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	6
Number of Observations Used	6

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	2.19615	2.19615	5.07	0.0875
Error	4	1.73233	0.43308		
Corrected Total	5	3.92848			

Root MSE	0.65809	R-Square	0.5590
Dependent Mean	30.05167	Adj R-Sq	0.4488
Coeff Var	2.18986		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	30.65667	0.37995	80.69	<.0001
x1	1	-1.21000	0.53733	-2.25	0.0875

Using PROC GLM. Skip dummy variables.

Using PROC GLM. Specify the factor levels in CLASS.

Using PROC GLM.

$$H_0 : \mu_1 - \mu_2 = [1, -1] \cdot \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = 0:$$

Using PROC GLM.

$$H_0 : \mu_1 - \mu_2 = [1, -1] \cdot \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = 0:$$

```
/* Test H0: mu_1 = mu_2 using PROC GLM */
PROC GLM DATA=study;
    CLASS Treatment;
    MODEL y = Treatment;
    CONTRAST 'Trt 1 vs Trt 2' Treatment 1 -1;
RUN;
QUIT;
```

UNEQUAL REPLICATION

Subsample nos	Subsample means
Treatment 1 (two day conditioning)	
6	29.53
7	29.90
Treatment 2 (three day conditioning)	
8	30.37
9	29.10
10	30.80

$$H_0 : \mu_1 - \mu_2 = [1, -1] \cdot \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = 0$$

So PROC REG and PROC GLM are unchanged.

Fitted model:

$$\begin{bmatrix} 29.53 \\ 29.90 \\ 30.37 \\ 29.10 \\ 30.80 \end{bmatrix} = \mu_1 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \mu_2 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} e_{11} \\ e_{12} \\ e_{21} \\ e_{22} \\ e_{23} \end{bmatrix}$$

Computing grand mean μ by hand changes:

$$\mu = \frac{\mu_1 + \mu_2}{2}$$

Computing grand mean μ by hand changes:

$$\hat{\mu} = \frac{\hat{\mu}_1 + \hat{\mu}_2}{2} = \frac{1}{2}\left(\frac{y_{11} + y_{12}}{2}\right) + \frac{1}{2}\left(\frac{y_{21} + y_{22} + y_{23}}{3}\right)$$

$$= \frac{1}{4}y_{11} + \frac{1}{4}y_{12} + \frac{1}{6}y_{21} + \frac{1}{6}y_{22} + \frac{1}{6}y_{23}$$